

## Part VII

### Intro to ML + Bayesian Decision Theory

# Lecture Outline

① Intro to Machine Learning

② Uncertainty

③ Probability Recab

④ Bayesian Decision Theory

# What is Machine Learning?

**Machine Learning:** is the field concerned with using *optimization* techniques to develop strategies that leverage *data* to optimize performance criteria.

## Examples

- Recognizing objects from pictures.
- Predicting the next word in a sentence.
- Projecting your data into linearly independent dimensions.



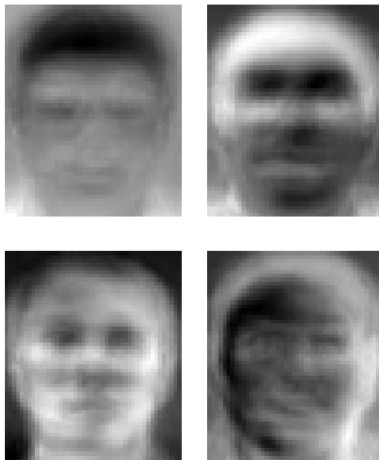
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Your favorite LLM.

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## Related Fields

Related fields to machine learning:

- **Statistics:** Statistical estimation is concerned with the same problem of machine learning. Most machine-learning strategies are statistical.
- **Optimization:** Machine learning is based on using optimization principles with data to optimize performance criteria.

# Types of Machine Learning

Types of Machine Learning:

- **Supervised:** given input data with output data (labels), predict the output for a test input data:
  - Regression (e.g., predict CO2 concentrations after two years).
  - Classification (e.g., cancer image classification).

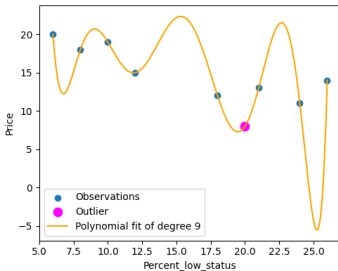
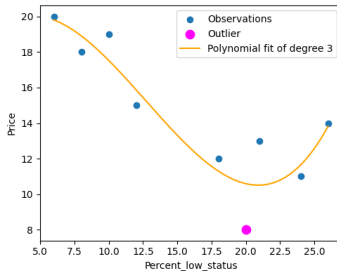
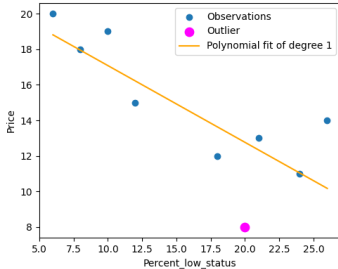
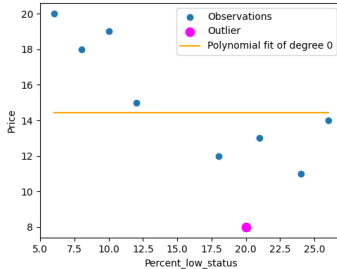
# Types of Machine Learning

## Types of Machine Learning:

- **Supervised:** given input data with output data (labels), predict the output for a test input data:
  - Regression (e.g., predict CO2 concentrations after two years).
  - Classification (e.g., cancer image classification).
- **Unsupervised Learning:** Given only input data (no output data), find a structure in the data.
  - Dimensionality reduction (e.g., projecting 4D data into 3D for visualization.).
  - Estimating probability density function (e.g., the density of your grades.).
  - Clustering (e.g., discover types of sleep patterns).

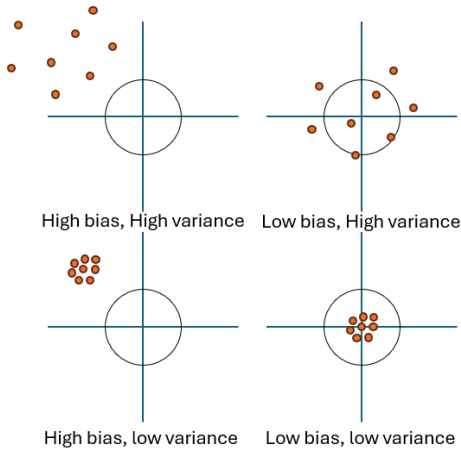
# Underfitting and Overfitting

## House Prices data



# Underfitting and Overfitting

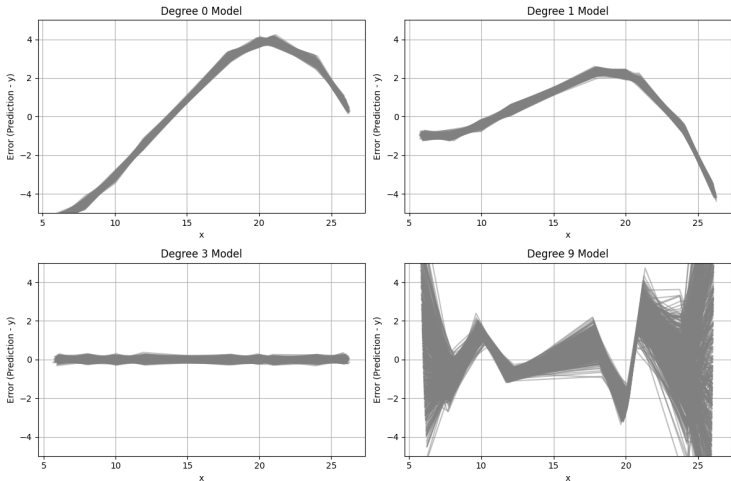
The Bias-Variance tradeoff.



# Underfitting and Overfitting

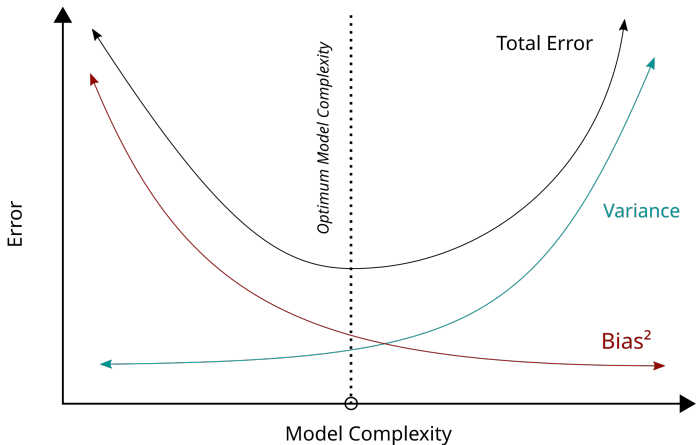
The Bias-Variance tradeoff.

We plot the errors with the polynomial models for different new data



## Underfitting and Overfitting

- There is always a trade-off between bias (underfitting) and variance (overfitting), with very flexible models having low bias and high variance, and relatively rigid models having high bias and low variance.
- This problem has not been solved completely yet!



## Underfitting and Overfitting

- There is always a trade-off between bias (underfitting) and variance (overfitting), with very flexible models having low bias and high variance, and relatively rigid models having high bias and low variance.
- This problem has not been solved completely yet!
- During optimization (training), divide your data into **training data, validation data, and test data**.
- **Trainig data** is used to fit your model. **Validation data** is used during training to evaluate if your model is overfitting or underfitting. **Test data** is used to evaluate your final model.

## Cross-Validation

**Cross-Validation:** A technique to evaluate model performance by rotating training/validation splits.

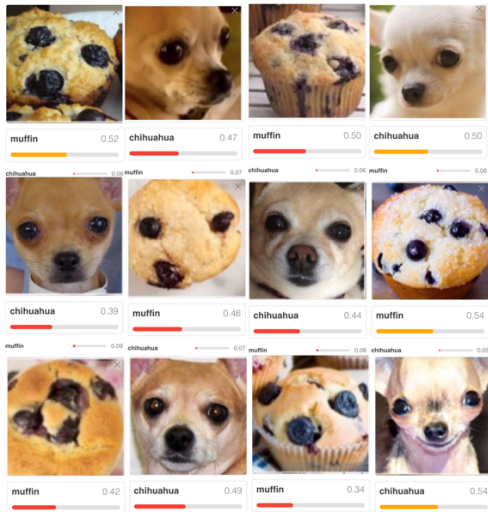
- Split data into training+validation part and testing part.
- **k-Fold CV:** Split training+validation data into  $k$  equal parts (folds). Train on  $k - 1$  folds, validate on the remaining fold. Repeat  $k$  times and average results.
- Reduces reliance on a single train/validation split.
- Helps estimate generalization error (errors based on data that was not used for training) more reliably.

# Uncertainty

**Uncertainty** is a key concept in any decision system. "You have disease A" is not the same as "You have a 55% chance of having disease A".

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# Uncertainty

Uncertainty arises due to several factors:

- Errors in the data (e.g., noisy sensor).
- Limited data.
- Inaccurate model assumptions.

# Uncertainty

**Probability theory** provides a consistent framework for quantizing and manipulating uncertainty.

## Probability with Events

We toss a *fair* coin with Heads ( $H$ ) and Tails ( $T$ ) 2 times. The set of possible outcomes is  $\Omega = \{(H, H), (H, T), (T, H), (T, T)\}$  and it is called the *Sample Space*.

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- $A$  : Event of the second toss being a head.  $P(A) = 1/2$  (fair coin).
- $B$  : Event of getting two heads or two tails.  $P(B) = 1/2$  (fair coin).
- $P(\Omega) = 1$

## Sum Rule, Product Rule, and Conditional Probability (Events)

### Conditional Probability:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad \text{if } P(B) \neq 0.$$

### Product Rule:

$$P(A \cap B) = P(A | B)P(B) = P(B | A)P(A).$$

**Independence** Two events  $A$  and  $B$  are **independent** if

$$P(A \cap B) = P(A)P(B).$$

Equivalently,  $P(A | B) = P(A)$  and  $P(B | A) = P(B)$ .

### Sum Rule (Law of Total Probability):

$$P(A) = P(A \cap B) + P(A \cap B^c) = P(A | B)P(B) + P(A | B^c)P(B^c)$$

$$\text{or more generally, } P(A) = \sum_i P(A \cap B_i) = \sum_i P(A | B_i)P(B_i),$$

for any partition  $\{B_i\}$  of  $\Omega$ .

## Medical Diagnoses Example

- Let  $D$  be the event that a patient has a specific disease.
- Let  $T$  be the event that a medical test is positive.

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- Let  $D$  be the event that a patient has a specific disease.
- Let  $T$  be the event that a medical test is positive.
- The disease prevalence (prior) is  $P(D) = 0.01$  (1 in every 100 has a disease).
- The test is 90% sensitive, so  $P(T | D) = 0.90$  (likelihood).
- The test has a 5% false positive rate, so  $P(T | D^c) = 0.05$ .

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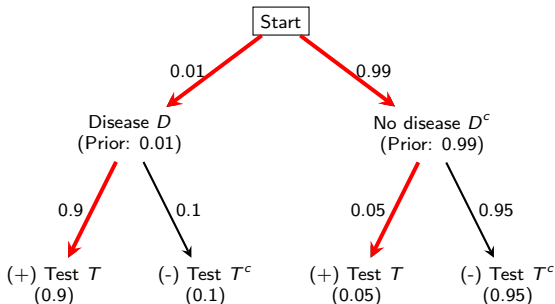


Figure: Probability Tree.

Then, the overall probability of a positive test (the probability of the evidence) is:

$$P(T) = P(T | D)P(D) + P(T | D^c)P(D^c) = 0.90 \cdot 0.01 + 0.05 \cdot 0.99 = 0.0585.$$

## Bayes' Rule (Events)

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B)}.$$

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}.$$

## Random Variable

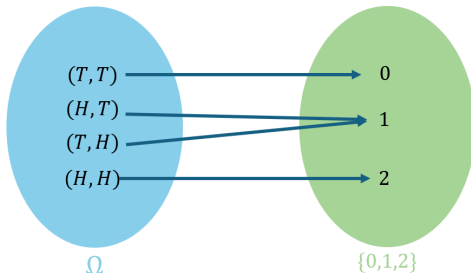
A **Random Variable** is a function which maps events in the sample space  $\omega \in \Omega$  to numbers in a chosen set:

$$X : \Omega \rightarrow \mathbb{X}$$

. Example:

- Define  $X$  is the number of heads:  
 $X((H, H)) = 2, X((H, T)) = 1, X((T, H)) = 1, X((T, T)) = 0$ .
- $P(X = 2) = P(X^{-1}(2)) = P((H, H)) = 1/4$ .
- $P(X = 1) = P(X^{-1}(1)) = P(\{(H, T), (T, H)\}) = 1/2$ .

$$X: \Omega \rightarrow \{0,1,2\}$$



## Random Variable

A random variable can also have *continuous values*.

- Define  $X : \Omega \rightarrow \mathbb{R}$  as the random variable representing temperatures in celsius.
- Define  $Y : \Omega \rightarrow \mathbb{R}$  as the random variable representing temperatures in Fahrenheit.
- $Y = \frac{5(X-32)}{9}$  (conversion from celsius to Fahrenheit).

## pdfs and pmfs (distributions)

- For a discrete random variable  $X : \Omega \rightarrow \mathbb{X}$  with a probability mass function (pmf)  $p_X(x)$ , we have

$$P(X = x) = p_X(x)$$

and

$$\sum_{x \in \mathbb{X}} p_X(x) = 1.$$

- For a continuous random variable  $X : \Omega \rightarrow \mathbb{X}$  with a probability density function (pdf)  $p_X(x)$ , we have

$$P(X = x) = 0,$$

$$P(X \in [x_1, x_2]) = P(x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} p_X(x) dx = \int_{x \in [x_1, x_2]} p_X(x) dx,$$

and

$$\int_{x \in \mathbb{X}} p_X(x) dx = 1.$$

## pdfs and pmfs (distributions)

- We write  $X \sim p_X(x)$  to say that  $X$  has a pdf/pmf (or distributed according to )  $p_X(x)$ .

## pdfs and pmfs (distributions)

Can you give examples of some common pdfs and pmfs?

## Probability Rules with pmfs/pdfs

**Joint distribution** of two random variables ( $X, Y$ ):

$$p_{X,Y}(x, y) = P(X = x, Y = y) \quad (\text{discrete}),$$

$$p_{X,Y}(x, y) \quad (\text{pdf for continuous}).$$

**Fundamental Rules** (mirroring the event rules):

- **Marginal (Sum/Integral) Rule (Total Law of Probability):**

$$p_X(x) = \sum_y p_{X,Y}(x, y) \quad (\text{discrete}), \quad p_X(x) = \int p_{X,Y}(x, y) dy \quad (\text{continuous}).$$

- **Conditional:**

$$p_{X|Y}(x | y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

- **Product Rule:**

$$p_{X,Y}(x, y) = p_{X|Y}(x | y) p_Y(y).$$

- **Bayes' Rule:**

$$p_{X|Y}(x | y) = \frac{p_{Y|X}(y | x) p_X(x)}{p_Y(y)}.$$

- **Independence:**

$$p_{X,Y}(x, y) = p_X(x) p_Y(y) \iff p_{X|Y}(x | y) = p_X(x).$$

## Expected Value

We call the average value of some function  $f(x)$  under a probability distribution  $p_X(x)$  the *expectation of  $f(x)$*  and will be defined as

$$\mathbb{E}[f(x)] = \sum_{x \in \mathbb{X}} p_X(x) f(x) \text{ (discrete), } \mathbb{E}[f(x)] = \left( \int_{x \in \mathbb{X}} p_X(x) f(x) dx \right) \text{ (continuous).}$$

We also consider the *conditional expectation* with respect to a conditional distribution  $p_{X|Y}(x | y)$  as

$$\mathbb{E}[f(x) | y] = \sum_{x \in \mathbb{X}} p_{X|Y}(x | y) f(x) \text{ (discrete),}$$
$$\mathbb{E}[f(x) | y] = \left( \int_{x \in \mathbb{X}} p_{X|Y}(x | y) f(x) dx \right) \text{ (continuous).}$$

# Bayesian Decision Theory: Overview

## Bayesian Decision Theory

provides a rigorous framework for making decisions under *uncertainty*, leveraging the probability distributions we have been discussing (e.g.,  $p_{Y|X}(y|x)$ ).

- **Possible Outcomes:** Denote them by  $Y \in \mathbb{Y}$  (e.g.,  $Y = 1, 2, \dots, K$  for  $K$  classes).
- **Measurements (Data):** Denote them by  $X \in \mathbb{X}$ , with a joint (or conditional) distribution  $p_{X,Y}(x,y) = p_{Y|X}(y|x)p_X(x)$ .
- **Actions (Decisions):** Denote by  $a \in \mathcal{A}$  (e.g., choose label 1, choose label 2,  $\dots$ , or “reject,” or “re-measure,” etc.).
- **Loss Function:**  $L(y, a)$  quantifies how costly it is to take action  $a$  when the true outcome is  $y$ .

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### Goal:

Choose the action  $a^*$  that *minimizes* the expected loss (risk) given the observed data.

## Posterior Distribution and Risk

**Posterior Distribution:** From our probability recap, once we observe  $X = x$ , we form the posterior distribution over the possible outcomes  $Y$  as

$$p_{Y|X}(y | x) = \frac{p_{X|Y}(x | y) p_Y(y)}{p_X(x)}.$$

**Expected Loss (Conditional Risk):**

$$R(a | x) = \sum_{y \in \mathbb{Y}} p_{Y|X}(y | x) L(y, a) \quad (\text{discrete } Y),$$

or

$$R(a | x) = \int_{y \in \mathbb{Y}} p_{Y|X}(y | x) L(y, a) dy \quad (\text{continuous } Y).$$

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**Bayesian Decision Rule:**

$$a^*(x) = \underset{a \in \mathcal{A}}{\operatorname{argmin}} R(a | x)$$

## 0–1 Loss Case: Maximum A Posteriori Rule

**Special Case: 0–1 Loss.** Suppose we have a set of outcomes  $\mathcal{C} = \{1, \dots, K\}$  representing classes and a corresponding set of actions  $\mathcal{A} = \{a_1, \dots, a_K\}$ , where  $a_i$  is the action “choose class  $c$  for  $c \in \{1, \dots, K\}$ ”. Define the loss function as:

$$L(c, a_i) = \begin{cases} 0, & \text{if } i = c \quad (\text{correct}), \\ 1, & \text{if } i \neq c \quad (\text{wrong}). \end{cases}, \quad i, c \in \mathcal{C}$$

Then the **expected loss** for choosing  $a_i$  given  $X = x$  is:

$$\begin{aligned} R(a_i | x) &= \sum_{c=1}^K p_{C|X}(c | x) L(c, a_i) = p_{C|X}(i | x) \cdot 0 + \sum_{\substack{c=1 \\ c \neq i}}^K p_{C|X}(c | x) \\ &= 1 - p_{C|X}(i | x) \quad (\text{since } \sum_{c=1}^K p_{C|X}(c | x) = 1). \end{aligned}$$

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**Bayesian Decision Rule:**

$$a^*(x) = \underset{a \in \{a_1, \dots, a_K\}}{\operatorname{argmin}} R(a | x) = a_{\operatorname{argmax}_{c \in \mathcal{C}} p_{C|X}(c | x)}$$

**Interpretation:** Under 0–1 loss, the optimal action is to pick the class  $c \in \mathcal{C}$  with the *highest* posterior probability. This is known as the **Maximum A Posteriori (MAP)** rule.

## Bayesian Decision-Making Depends on Proportionality:

$$\underbrace{p_{C|X}(c | x)}_{\text{Posterior}} \propto \underbrace{p_{X|C}(x | c)}_{\text{Likelihood}} \times \underbrace{p_C(c)}_{\text{Prior}}.$$

- The denominator  $p_X(x)$  (evidence) is a **normalization constant**—it does not depend on  $c$ .
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$$a^*(x) = a_{\operatorname{argmax}_{c \in \mathcal{C}} p_{C|X}(c | x)} = a_{\operatorname{argmax}_{c \in \mathcal{C}} p_{X|C}(x | c) p_C(c)}$$

## Incorporating a Reject Option

**Motivation:** In critical applications (e.g., medical diagnosis), abstaining from uncertain decisions can reduce costly errors.

**Extended Action Space:** Add  $a_{\text{reject}}$  with loss  $\lambda$ , where  $0 \leq \lambda < 1$ .

**Loss Function for classification:**

$$L(c, a_i) = \begin{cases} 0 & \text{if } i = c \quad (\text{correct}), \\ 1 & \text{if } i \neq c \quad (\text{wrong}), \\ \lambda & \text{if } a_i = a_{\text{reject}} \quad (\text{reject}). \end{cases}$$

**Expected Risks:**

- For action  $a_i$ :  $R(a_i | x) = 1 - p_{C|X}(i | x)$
- For  $a_{\text{reject}}$ :  $R(a_{\text{reject}} | x) = \sum_{c=1}^K p_{C|X}(c | x)\lambda = \lambda$

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**Decision Rule:**

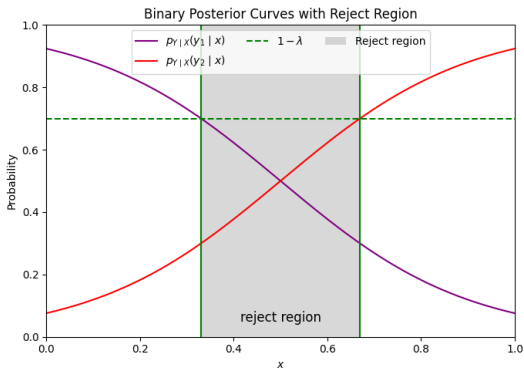
$$a^*(x) = \begin{cases} a_{\arg\max_{c \in C} p_{C|X}(c | x)} & \text{if } \max_{c \in C} p_{C|X}(c | x) > 1 - \lambda, \\ a_{\text{reject}} & \text{otherwise.} \end{cases}$$

## Example: Reject Threshold in Binary Classification

**Scenario:** Binary classification ( $C \in \{0, 1\}$ ) with posterior  $p_{C|X}(c | x)$ .

### Threshold Analysis:

- If  $\lambda = 0.3$ : Threshold =  $1 - \lambda = 0.7$ .
- Classify as 1 if  $p_{C|X}(1 | x) > 0.7$ , 0 if  $p_{C|X}(0 | x) > 0.7$  ( $p_{C|X}(1 | x) < 0.3$ ), else reject.



## Example: Medical Testing

### Scenario:

- $C \in \{0, 1\}$ . with 0 being for the event that "Disease is present" and 1 is for the event that "Disease is not present."
- We measure  $X$ : some test indicator (e.g., a biomarker level) with distribution  $p_{X|C}(x | c)$ , or some machine learning model which takes data and returns a diagnosis with a distribution  $p_{X|C}(x | c)$ .
- Actions  $\mathcal{A} = \{\text{Treat, Do not treat, Reject}\}$ .
- Loss function  $L(c, a)$  might look like:
  - $L(\text{Disease Present, Treat})$  is relatively small (the cost of medication).
  - $L(\text{Disease Present, Do not treat})$  is very large (risk of leaving a severe condition untreated).
  - $L(\text{No Disease, Treat})$  is moderate (unnecessary side effects, wasted resources).
  - $L(\text{No Disease, Do not treat})$  is small (correctly avoiding unnecessary treatment).
  - $L(y, \text{Reject})$  includes the cost of deferring the decision or ordering additional tests.

### Bayesian Approach:

- Obtain the posterior  $p_{C|X}(c | x)$  from prior knowledge  $p_C(c)$  and likelihood  $p_{X|C}(x | c)$  of the test results for data or datasheet.
- Compute the expected loss for each action and pick the action that has a minimal expected loss.